FEAT-ZD-48600 · BE-05 Coding · Code changes on the working branch

Backend agent · item task.code · source hobasa-backend-agent/outputs/FEAT-ZD-48600/task.code.md · approved by automated checks and independent judge

task.code: FEAT-ZD-48600

BE-05 "Coding". This file records the code changes made in ap_aging for TSK-05 (STORY-05) and TSK-06 (STORY-06), as task.components lists them. The changes are in the working tree of `feature/story` and are not committed. The spine commits and pushes them after evaluation.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	BE-05 Coding	record:brief
Owner / gate	Developer; automated checks, the independent judge and sampled human evaluation	record:brief
Inputs	task.components, feat.design, feat.stories, org.engineering_stack, org.lessons; task.secrets_scope by reference only	record:brief
Background read	feat.clarifications (DES-03)	journey:feat.clarifications
Repository / branch	ap_aging, `feature/story`	record:brief; task.components
Write scope of this step	`src/ap_aging/`, `docs/`, `README.md`	record:brief
Stories / tasks	STORY-05 / TSK-05 (`overdue_with_risk`); STORY-06 / TSK-06 (`overdue_rows_to_json`)	task.components; feat.stories

1. Files changed

File	Change	Task	Source
`src/ap_aging/aging.py`	Additive. Adds `import json`, `List` in the typing import, the constants `OVERDUE_AFTER_DAYS` and `TWO_PLACES`, and the functions `overdue_with_risk` and `overdue_rows_to_json`	TSK-05, TSK-06	tool:repo
`src/ap_aging/__init__.py`	Imports and exports the two new functions; `__version__` is `"1.5.0"`	TSK-05, TSK-06	tool:repo
`README.md`	Lists the three public functions and links to `docs/ap_aging.md`	Documentation	file:ap_aging/README.md
`docs/ap_aging.md`	New module overview: the invoice input, then each public function with its signature, inputs, outputs, rules, errors and examples, then the helpers and constants	Documentation	file:ap_aging/docs/ap_aging.md

Not changed: `tests/` (`tests/__init__.py` and `tests/test_aging.py`), which is outside this step's write scope, and `ci.py` (tool:repo; record:brief).

2. What was implemented

2.1 `src/ap_aging/aging.py` *(task.components A1 to A6)*

#	Change in the code	Matches	Source
A1	`import json`	feat.design B3.1	task.components; tool:repo
A2	`from typing import Dict, Iterable, List` (the existing line, extended)	feat.design B3.1	task.components; tool:repo
A3	`OVERDUE_AFTER_DAYS = 30`, under a comment saying an invoice exactly 30 days past due is not listed	feat.design B3.2, DD-4	task.components; tool:repo
A4	`TWO_PLACES = Decimal("0.01")`	feat.design B3.2	task.components; tool:repo
A5	`overdue_with_risk(invoices: Iterable[Dict], as_of: date, risk_by_vendor: Dict[str, str]) -> List[Dict]`	feat.design B3.2 steps 1 to 4	task.components; tool:repo
A6	`overdue_rows_to_json(rows: Iterable[Dict]) -> str`	feat.design B3.2 (CT-1)	task.components; tool:repo

`overdue_with_risk` works in four steps (feat.design B3.2; tool:repo):

1 It skips an invoice when `inv.get("paid")` is truthy. `aging_buckets` uses the same test (D-2026-050).

2 It computes `days = days_past_due(inv["due_date"], as_of)`, reusing `days_past_due` unchanged (calendar days,

D-2026-031). It skips the invoice when `days <= OVERDUE_AFTER_DAYS`, so only invoices more than 30 days past due are kept and an invoice exactly 30 days past due is not listed. It never calls `bucket_for`. An inline comment says why: `bucket_for(30)` returns `"31-60"`.

1 It builds a row with exactly the keys `id`, `vendor`, `days`, `amount` and `risk`:

2 `amount` is `Decimal(inv["amount"]).quantize(TWO_PLACES)`;

3 `risk` is `risk_by_vendor.get(inv["vendor"], "unknown")`;

4 `vendor` is read by indexing (`inv["vendor"]`), so a missing `vendor` on an overdue unpaid invoice raises

`KeyError`.

1 It sorts with `rows.sort(key=lambda row: (-row["days"], row["id"]))` and returns the list.

`as_of` is a required argument with no default. The function builds a new list of new row dicts and does not modify its inputs (tool:repo; feat.design NF-10).

`overdue_rows_to_json` is a single `json.dumps` call over a list comprehension (tool:repo; feat.design B3.2). Each object has its keys in CT-1 order:

- `id`;
- `vendor`;
- `days`, as `int(r["days"])`;
- `amount`, as `str(Decimal(r["amount"]).quantize(TWO_PLACES))`;
- `risk`.

Rows keep their order, and no date field is added. Neither function uses `float` (tool:repo; record:memory D-2026-044; org.lessons L-011).

Unchanged: `BUCKETS`, `days_past_due`, `bucket_for` and `aging_buckets` keep the signatures and bodies that task.components section 1 records (tool:repo; task.components; AC-05.9).

Comments and docstrings (tool:repo):

- The `overdue_with_risk` docstring states the row keys, the sort order, the `'unknown'` default for a vendor missing

from `risk_by_vendor`, the `KeyError` for a missing `vendor`, and that the function is pure.

- The `overdue_rows_to_json` docstring states the CT-1 shape, the `'[]'` result for no rows, that rows keep their

order, and that the amount never passes through `float`.

- The comment above `OVERDUE_AFTER_DAYS` says an invoice is overdue only when strictly more than this many calendar

days past due, so exactly 30 is not listed.

- The comment above the threshold check in `overdue_with_risk` explains why `bucket_for` is not used:

`bucket_for(30)` is `"31-60"`, but an invoice exactly 30 days past due must not be listed.

2.2 `src/ap_aging/__init__.py` *(task.components I1 to I3)*

```
from .aging import aging_buckets, overdue_rows_to_json, overdue_with_risk

__version__ = "1.5.0"
__all__ = ["aging_buckets", "overdue_with_risk", "overdue_rows_to_json"]
```

The version takes one minor step, from `"1.4.0"` (task.components section 1) to `"1.5.0"` (tool:repo). feat.design DD-11 and feat.clarifications CL-X8 call for a minor step "for the lane to confirm", but no input gives the number (task.components BC-2). The Developer confirms it at review (open question Q-2).

2.3 Documentation

`docs/ap_aging.md` (new) covers (file:ap_aging/docs/ap_aging.md):

- a module overview;
- the DC-1 invoice keys;
- each public function with its signature, inputs, rules, output and errors;
- examples that use only synthetic data (`SYN-INV-*`, "Synthetic Vendor <letter>");
- a table of the module's helpers and constants.

`README.md` gives a one-line summary of each public function, links to the docs, and keeps the CI command (file:ap_aging/README.md).

task.components section 2 had listed `README.md` as unchanged. This step's brief asks for README.md and docs/ to be updated, so the brief was followed (record:brief; task.components).

3. Traceability to acceptance criteria

AC	Implemented by	Test to add in BE-06
AC-05.1 (31 days is listed; `days` comes from `days_past_due`)	Step 2 of `overdue_with_risk`	`test_overdue_boundary_30_and_31`

AC	Implemented by	Test to add in BE-06
AC-05.2 (exactly 30 days is not listed)	The `days <= OVERDUE_AFTER_DAYS` skip; no call to `bucket_for`	`test_overdue_boundary_30_and_31`
AC-05.3 (paid invoices are not listed)	Skip on `inv.get("paid")`	`test_overdue_excludes_paid_and_not_due`
AC-05.4 (invoices not yet due are not listed)	`days_past_due` returns 0 when the invoice is not yet due	`test_overdue_excludes_paid_and_not_due`
AC-05.5 (exactly five keys)	The row literal	`test_overdue_row_keys_and_sort`
AC-05.6 [Pending PQ-2] (days descending, then id ascending)	`rows.sort(...)`	`test_overdue_row_keys_and_sort`
AC-05.7 (`'unknown'` risk)	`risk_by_vendor.get(..., "unknown")`	`test_overdue_unknown_risk`
AC-05.8 (`Decimal("980.5")` becomes `Decimal("980.50")`)	`quantize(TWO_PLACES)`	`test_overdue_amount_quantised`
AC-05.9 (existing functions unchanged; no call to `bucket_for`)	Additive change only	`test_existing_aging_unchanged`
AC-05.10 [Pending TQ-9] (`KeyError` for a missing vendor)	`inv["vendor"]`	`test_overdue_missing_vendor_raises`
AC-06.1 to AC-06.4 [Pending TQ-3] (CT-1 JSON, `"1250.00"`, `[]`, no date field)	`overdue_rows_to_json`	`test_rows_to_json`

Sources: feat.stories (STORY-05, STORY-06); task.components (section 3.3); tool:repo.

4. Quality bar and test status

Item	Status	Source
CI command	`/usr/bin/python3 ci.py`	org.engineering_stack; file:ap_aging/README.md
CI gates	Compile, unit tests, coverage of at least `COVERAGE_MIN = 90.0`, static checks	file:ap_aging/ci.py; org.engineering_stack
Tests in the repository	`tests/test_aging.py` imports only `aging_buckets`, `bucket_for` and `days_past_due`. No test calls `overdue_with_risk` or `overdue_rows_to_json`	tool:repo
Effect on the coverage gate	The two new functions are not exercised until BE-06 adds the tests in task.components section 3.3, so the coverage gate depends on BE-06 (open question Q-1)	tool:repo; task.components; file:ap_aging/ci.py
Static-check rules followed in the new code	No `print`, `eval`, `exec` or bare `except`; Python 3.9 typing only (`Dict`, `Iterable`, `List`)	tool:repo; feat.design (Part B conventions)

5. Decisions and lessons applied

Item	How	Source
D-2026-031: calendar days	`days_past_due` is reused	record:memory; tool:repo
D-2026-044 and L-011: Decimal, never float	`Decimal(...).quantize(TWO_PLACES)`; the JSON amount is made with `str()`	record:memory; org.lessons; tool:repo
D-2026-050: paid invoices excluded	Skip on `inv.get("paid")`	record:memory; tool:repo
L-007: test the boundary day	Strict threshold on 30; the boundary test is listed for BE-06	org.lessons; task.components
Keep existing public signatures; add new functions	No existing signature changed	file:ap_aging/CLAUDE.md; tool:repo

Item	How	Source
No secrets or personal data	None used; examples use synthetic data	feat.design (SE-4); file:ap_aging/docs/ap_aging.md

6. Open questions (non-blocking)

ID	Question	Interim handling	Owner
Q-1	The CI coverage gate (90.0) is not met at the end of BE-05, because the new functions have no tests yet and `tests/` is outside this step's write scope. Is it accepted that the gate is met at BE-06?	BE-06 adds the eight tests in task.components 3.3 and re-runs CI	Developer
Q-2	Confirm the ap_aging version `1.5.0`, a minor step from `1.4.0` (task.components BC-2)	`1.5.0` is set	Developer
—	Carried unchanged: PQ-2 (AC-05.6), TQ-9 (AC-05.10), TQ-3 (AC-06.1 to AC-06.4), TQ-1 (`bucket_for(30)`), DQ-2 (rounding mode), CQ-6 (float guard), RA-1 (shared branch)	As recorded in the source files	As recorded

Tool results (recorded by the spine from the real tools)

[Tool result] [ap_aging] Sources compile